

1. $I = 1.92 \text{ A} \therefore \frac{1}{R_p} = \frac{1}{R_1} + \dots + \frac{1}{R_n}, V = IR$
2. $d = 1.89 \times 10^{-6} \text{ m} \therefore P = \frac{V^2}{R}, R = \frac{\rho L}{A} \text{ and } A = \pi r^2$
3. $R_{eq} = 1.50 \Omega \therefore R_s = R_1 + R_2 + R_3, \frac{1}{R_p} = \frac{1}{R_1} + \dots + \frac{1}{R_n}$
4. $\dot{N}_e = 3.72 \times 10^{20} \text{ electrons per second} \therefore \Delta V = \frac{\Delta PE}{q}, W_{c.f.} = \Delta PE \therefore W = -q\Delta V \therefore \frac{q}{t} = \frac{-W}{t\Delta V} \therefore \dot{N}_e = \frac{q}{qe t} = \frac{-W}{qe t \Delta V}$
5. $\theta_2 = 4.70^\circ \therefore n_1 \sin \theta_1 = n_2 \sin \theta_2$
6. $\frac{F}{\ell} = 170. \frac{\text{N}}{\text{m}} \therefore F = I\ell B \sin \theta, \theta = 90^\circ$
7. $f = 133.2 \text{ Hz} \therefore f_{obs} = f_s \left(\frac{v_w}{v_w \pm v_s} \right)$ (review the Doppler effect)
8. $Q_{cloud} = 351. \text{ C} \therefore C = \epsilon_0 \frac{A}{d}, \Delta V = Ed, Q = CV$
9. $q = 8.28 \times 10^{-9} \text{ C} \therefore \Sigma \vec{F}_x = m\vec{a}_x, F = k \frac{|q_1 q_2|}{r^2}$
10. $E = 2.63 \times 10^4 \text{ J} \therefore P = IV = \frac{E}{t}$
11. $T = 1.56 \text{ ms} \therefore T = \frac{1}{f}$
12. $F = 4.25 \times 10^{-5} \text{ N} \therefore F = qvb \sin \theta \therefore F_{max} = qvb$
13. $\theta = 72.7^\circ \therefore I = I_0 \cos^2 \theta$
14. $P = 65.1 \text{ D} \therefore \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}, P = \frac{1}{f}$
15. $d = 1.93 \text{ m} \therefore \Delta E_t = W_{nc} = -f_k d, f_k = \mu_k N = \mu_k mg, PE = \frac{1}{2} kx^2$
16. $\text{emf} = 20.6 \text{ V} \therefore \text{emf} = -N \frac{\Delta \Phi}{\Delta t}$
17. $E_{ind} = 0.0865 \text{ J} \therefore E_{ind} = \frac{1}{2} LI^2$
18. $I_{RMS} = 1.98 \times 10^3 \text{ A} \therefore P_{ave} = I_{RMS} V_{RMS}$
19. $f = 3.55 \times 10^3 \text{ Hz} \therefore f_n = n \frac{v_w}{4L}, v_w = \left(331 \frac{\text{m}}{\text{s}} \right) \sqrt{\frac{T}{273 \text{ K}}}$
20. $C_T = 2.48 \times 10^{-9} \text{ F} \therefore \frac{1}{C_s} = \frac{1}{C_1} + \dots + \frac{1}{C_n}, C_p = C_1 + \dots + C_n$
21. $P = 24.4 \text{ D} \therefore P = \frac{1}{f}$
22. $v = 1.67 \times 10^8 \frac{\text{m}}{\text{s}} \therefore n = \frac{c}{v}$
23. $m = -0.556 \therefore \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}, \frac{h_i}{h_0} = -\frac{d_i}{d_0} = m$
24. $\epsilon = 1.23 \times 10^4 \text{ V} \therefore \epsilon = B\ell v$
25. $\theta = 0.0422 \text{ arcseconds} \therefore \theta = 1.22 \frac{\lambda}{D}$
26. $B = 2.14 \times 10^{-4} \text{ T} \therefore r = \frac{mv}{qB}, m_e = 9.11 \times 10^{-31} \text{ kg}, q_e = 1.60 \times 10^{-19} \text{ C}$
27. $B_{max} = 6.33 \times 10^{-6} \text{ T} \therefore c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{E}{B} = f\lambda$
28. $Q = 1.71 \times 10^{-6} \text{ C} \therefore V = \frac{kQ}{r}$
29. $P = 65.0 \text{ W} \therefore P = IV$
30. $\theta_c = 50.3^\circ \therefore n_1 \sin \theta_1 = n_2 \sin \theta_2$

[illegible]

Name: _____

This final exam covers Chapters 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. Suppose the voltage supplied to a circuit is 3.2 V. There are three resistors wired in parallel: $R_1 = 6.3\,\Omega$, $R_2 = 3.0\,\Omega$, and $R_3 = 9.3\,\Omega$. Calculate the current leaving the battery.

$$V = IR \rightarrow I = \frac{V}{R_p} = \frac{3.2 \text{ V}}{(6 \cdot 3^{-1} + 3^{-1} + 9 \cdot 3^{-1})^{-1} \frac{\text{V}}{\text{A}}} = 1.919 \text{ A}$$

ANSWER: 1.919 A ✓

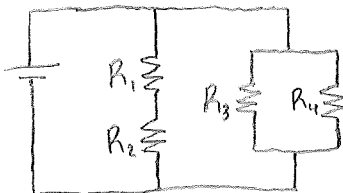
2. An incandescent lightbulb has a 1.2-cm-long filament and operates at 117. V to deliver 57. W. Calculate the diameter of the bulb's filament, assuming it to be a cylindrical tungsten wire, given that the resistivity of tungsten is $5.6 \times 10^{-8} \Omega \cdot \text{m}$.

$$P = \frac{V^2}{R} \rightarrow R = \frac{V^2}{P} = \frac{\rho L}{A} = \frac{\rho L}{\pi r^2} = \frac{4\rho L}{\pi d^2} \rightarrow d = \sqrt{\frac{4\rho L P}{V^2 \pi}}$$

$$d = \frac{4(5.6 \times 10^{-8} \frac{\text{V} \cdot \text{m}}{\text{A}})(0.012 \text{ m})(57 \text{ A} \cdot \text{V})}{(117 \text{ V})^2 \pi} = 1.888 \times 10^{-6} \text{ m}$$

ANSWER: $1.888 \times 10^{-6} \text{ m}$

3. A circuit has two branches connected in parallel: branch A consists of resistors $R_1 = 1.6\Omega$ and $R_2 = 6.5\Omega$ in series; branch B consists of resistors $R_3 = 8.0\Omega$ and $R_4 = 2.4\Omega$ in parallel. Draw this circuit and find the total equivalent resistance. Note: you must draw the circuit correctly to receive credit.



$$R_{in} = R_s = 1.6 \, \Omega + 6.5 \, \Omega = 8.1 \, \Omega$$

$$R_{24} = R_p = (8^{-1} + 2.4^{-1})^{-1} \Omega = 1.846 \Omega$$

$$R_T = R_p = (8.1^{-1} + 1.846^{-1})^{-1} \Omega = 1.503 \Omega$$

ANSWER: 1.503 Ω

+3

4. When a 2.0-V battery runs a single 119.-W light, how many electrons pass through it each second?

ANSWER: 3.719×10^{20} electrons/s

$$P = IV \rightarrow I = \frac{P}{V} = \frac{119 \frac{\text{C} \cdot \text{V}}{\text{A} \cdot \text{s}}}{2 \text{ V}} = 59.5 \frac{\text{C}}{\text{s}}$$

$$\frac{59.5 \frac{\text{C}}{\text{s}}}{1.6 \times 10^{-19} \frac{\text{C}}{\text{e}^-}} = 3.719 \times 10^{20} \text{ e}^-/\text{s}$$

5. Suppose that you would like to find the angle of refraction in a material. You shine a laser through air at an incident angle of 8.0° into the material which has an index of refraction of 1.7. Calculate the angle of refraction in the material.

ANSWER: 4.696°

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \rightarrow \theta_2 = \sin^{-1} \left(\frac{n_1 \sin \theta_1}{n_2} \right)$$

$$\theta_2 = \sin^{-1} \left(\frac{1 \sin(8^\circ)}{1.7} \right) = \boxed{4.696^\circ}$$

6. A wire carries a 71. A current toward the south. If a large magnet were placed directly to the west of the wire and emits a 2.4T magnetic field perpendicularly towards the wire, calculate the force exerted on the wire per length of wire.

ANSWER: $170.4 \frac{\text{N}}{\text{m}}$

$$F = ILB \sin \theta \rightarrow \frac{F}{L} = IB \sin \theta = (71 \text{ A})(2.4 \frac{\text{N}}{\text{A} \cdot \text{m}}) \sin(90^\circ) = \boxed{170.4 \frac{\text{N}}{\text{m}}}$$

7. Suppose a train sounding its 126.-Hz horn is moving at $17.8 \frac{\text{m}}{\text{s}}$. What frequency does a person standing by the tracks hear as the train approaches? Given this particular day's temperature, the speed of sound is $331. \frac{\text{m}}{\text{s}}$.

ANSWER: 133.161 Hz

$$f_{\text{obs}} = f_s \left(\frac{v_w}{v_w - v_s} \right) = (126 \text{ Hz}) \left(\frac{331 \text{ m/s}}{331 \text{ m/s} - 17.8 \text{ m/s}} \right) = \boxed{133.161 \text{ Hz}}$$

+4

8. Dry air will support electric fields up to its dielectric strength, $3.0 \times 10^6 \frac{\text{V}}{\text{m}}$. Above that value, an electric field creates enough ionization in the air to make the air a conductor, which causes a discharge or spark to reduce the field. Calculate the total charge on the charged region at the base of a thundercloud just before electrical breakdown occurs if 1.7 km of dry air separates the cloud from the ground. Model the cloud and ground as two parallel conducting plates and assume a thundercloud diameter of 4.1 km.

ANSWER: _____

$$\Delta V = E d = \frac{k |Q| d}{r^2} \rightarrow$$

1

9. Two identical charged spheres, each with a mass of 31. g, hang in equilibrium. They are suspended by a string of length 9.2 cm and are separated by a horizontal distance of 7.2 mm. Calculate the magnitude of the charge on each sphere.

ANSWER: _____

2

10. A 12.7-V car battery supplies a starter motor drawing 610 A for 3.4 s. Calculate the electrical energy delivered by the battery during this time.

$$IV = \frac{E}{t} \rightarrow E = t IV = (3.4 \text{ s})(610 \text{ A})(12.7 \text{ V}) = \boxed{26339.8 \text{ J}}$$

ANSWER: 26339.8 J

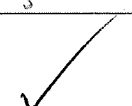
$$S.A.U = \frac{8.6 \cdot J}{8.6}$$



11. A cello generates a note with a pitch of 641. Hz. What is the period of the string's oscillation?

$$f = \frac{1}{T} \rightarrow T = \frac{1}{f} = \frac{1}{641 \frac{1}{s}} = \boxed{1.560 \times 10^{-3} \text{ s}}$$

ANSWER: $1.560 \times 10^{-3} \text{ s}$



+2

12. What is the magnitude of the maximum force exerted on an aluminum rod with a $9.2\text{-}\mu\text{C}$ charge that you pass between the poles of a 4.2-T permanent magnet at a speed of $1.1 \frac{\text{m}}{\text{s}}$?

$$F = qvb \sin \theta = (9.2 \times 10^{-6} \text{ C})(1.1 \frac{\text{m}}{\text{s}})(4.2 \frac{\text{N} \cdot \text{s}}{\text{C} \cdot \text{m}}) \sin(90^\circ) = 4.2504 \times 10^{-5} \text{ N}$$

ANSWER: $4.2504 \times 10^{-5} \text{ N}$

13. What angle would the axis of a polarizing filter need to make with the direction of polarized light of intensity 2.5 kW/m^2 to reduce the intensity to 0.22 kW/m^2 ?

$$I = I_0 \cos^2 \theta = I_0 (\cos \theta)^2 \rightarrow \cos \theta = \sqrt{\frac{I}{I_0}} \rightarrow \theta = \cos^{-1} \sqrt{\frac{I}{I_0}} = \cos^{-1} \sqrt{\frac{0.22 \text{ kW/m}^2}{2.5 \text{ kW/m}^2}} = 72.744^\circ$$

ANSWER: 72.744°

14. Calculate the power of the eye when viewing an object 8.0 cm away. Take the lens-to-retina distance to be 1.9 cm .

$$P = \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{0.08 \text{ m}} + \frac{1}{0.019 \text{ m}} = 65.132 \text{ D}$$

ANSWER: 65.132 D

15. Suppose a 0.315-kg object is connected to a horizontal spring. There is friction between the object and the surface, and the coefficient of kinetic friction μ_k is 0.218 . What total distance does the object travel if it is released from rest at a distance of 25 cm from its equilibrium position? The force constant of the spring is $41.7 \frac{\text{N}}{\text{m}}$.

ANSWER: _____

3

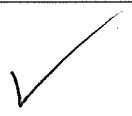
+3

16. Suppose a 130-turn coil lies in the plane of the page in a uniform magnetic field that is directed into the page. The coil originally has an area of 0.66 m^2 . It is then stretched for 0.79 s until it has no area. What is the magnitude of the induced emf if the uniform magnetic field has a strength of 0.19 T ?

ANSWER: 20.635 V

$$\mathcal{E} = \frac{-N\Delta\Phi}{\Delta t} = \frac{-N\Delta(BA\cos\theta)}{\Delta t} = \frac{-NB\cos\theta(A_f - A_i)}{\Delta t}$$

$$\mathcal{E} = \frac{-(130)(0.19 \frac{\text{V}\cdot\text{s}}{\text{m}\cdot\text{m}})\cos(0^\circ)(0 \text{ m}^2 - 0.66 \text{ m}^2)}{0.79 \text{ s}} = \boxed{20.635 \text{ V}}$$

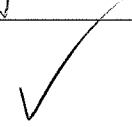


17. How much energy is stored in a 0.18-mH inductor when a 31-A current flows through it?

ANSWER: 0.08649 J

$$E_{\text{ind}} = \frac{1}{2}LI^2 = \frac{1}{2}(0.00018 \frac{\text{V}\cdot\text{s}}{\text{A}})(31 \text{ A})^2 = \boxed{0.08649 \text{ J}}$$

$$\text{V}\cdot\text{s}\cdot\text{A} = \frac{\text{J}\cdot\text{s}\cdot\cancel{\text{C}}}{\cancel{\text{C}}\cdot\text{s}}$$



18. What current flows through a high-voltage transmission line to transmit 950 MW of power at 480 kV ?

ANSWER: 1979.167 A

$$P = IV \rightarrow I = \frac{P}{V} = \frac{950 \times 10^6 \text{ A}\cdot\text{V}}{480 \times 10^3 \text{ V}} = \boxed{1979.167 \text{ A}}$$



19. Suppose you have a tube that is closed at one end. Its length is 21 cm and the air temperature is 0.17°C . What is the frequency of its fourth overtone?

ANSWER: 3548.506 Hz

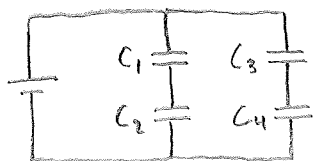
$$f_n = \frac{n v_w}{4L} = \frac{n(331 \frac{\text{m}}{\text{s}})\sqrt{\frac{T}{273 \text{ K}}}}{4L} = \frac{9(331 \frac{\text{m}}{\text{s}})\sqrt{\frac{(0.17 + 273.15) \text{ K}}{273 \text{ K}}}}{4(0.21 \text{ m})} = \boxed{3548.506 \text{ Hz}}$$

(correct)

4

+3

20. A circuit has two branches: one branch has capacitors 1 and 2 while the second branch has capacitors 3 and 4. Capacitors 1 and 2 are connected in series and capacitors 3 and 4 are connected in series. The two branches are connected in parallel. Draw this circuit and find the total capacitance given that $C_1 = 14 \text{ nF}$, $C_2 = 1.7 \text{ nF}$, $C_3 = 1.3 \text{ nF}$, and $C_4 = 3.8 \text{ nF}$. Note: you must draw the circuit correctly to receive credit.

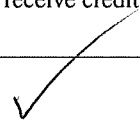


$$C_{12} = C_s = (14^{-1} + 1.7^{-1})^{-1} \text{ nF} = 1.516 \text{ nF}$$

$$C_{34} = C_s = (1.3^{-1} + 3.8^{-1})^{-1} \text{ nF} = 0.969 \text{ nF}$$

$$C_T = C_p = 1.516 \text{ nF} + 0.969 \text{ nF} = \boxed{2.485 \text{ nF}}$$

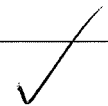
ANSWER: 2.485 nF



21. Suppose that you take a magnifying glass out on a sunny day and you find that it concentrates sunlight to a small spot 4.1 cm away from the lens. What is the power of the lens?

$$P = \frac{1}{f} = \frac{1}{0.041 \text{ m}} = \boxed{24.390 \text{ D}}$$

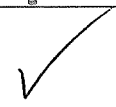
ANSWER: 24.390 D



22. Calculate the speed at which light travels in a particular material given that the index of refraction of the material is 1.8.

$$n = \frac{c}{v} \rightarrow v = \frac{c}{n} = \frac{3 \times 10^8 \text{ m/s}}{1.8} = \boxed{1.667 \times 10^8 \frac{\text{m}}{\text{s}}}$$

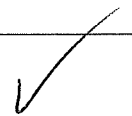
ANSWER: $1.667 \times 10^8 \frac{\text{m}}{\text{s}}$



23. A clear-glass light bulb is placed 0.42 m from a thin convex lens that has a focal length of 0.15 m. Calculate the magnification of the produced image.

$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f} \rightarrow d_i = \frac{1}{\frac{1}{f} - \frac{1}{d_o}}$$

ANSWER: -0.556



$$m = \frac{-d_i}{d_o} = \frac{-1}{d_o \left(\frac{1}{f} - \frac{1}{d_o} \right)} = \frac{-1}{(0.42 \text{ m}) \left(\frac{1}{0.15 \text{ m}} - \frac{1}{0.42 \text{ m}} \right)} = \boxed{-0.556}$$

+9

24. Calculate the motional emf induced along a 39.-km-long conducting object moving at an orbital speed of $8.5 \frac{\text{km}}{\text{s}}$ perpendicular to Earth's $37\text{-}\mu\text{T}$ magnetic field.

ANSWER: 12265.5 V

$$\mathcal{E} = Blv = (37 \times 10^{-6} \frac{\text{V}\cdot\text{s}}{\text{m}\cdot\text{T}}) (39 \times 10^3 \text{ m}) (8.5 \times 10^3 \frac{\text{m}}{\text{s}}) = \boxed{12265.5 \text{ V}}$$

25. The primary mirror of an orbiting space telescope has a diameter of 4.0m. One of the key advantages of operating in space is that it avoids atmospheric distortion, allowing the telescope to approach its diffraction-limited resolution. Assuming the telescope observes visible light with a wavelength of 670nm, what is the minimum angular separation between two point sources that can be just resolved according to the Rayleigh criterion? Report your answer in arcseconds. Note: 1 degree is 60 arcminutes and 1 arcminute is 60 arcseconds.

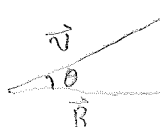
ANSWER: 0.0422 arcseconds

$$\theta = \frac{1.22\lambda}{D} = \frac{1.22(670 \times 10^{-9} \text{ m})}{4 \text{ m}} = 2.0435 \times 10^{-7} \text{ rad}$$

$$\frac{2.0435 \times 10^{-7} \text{ rad}}{\pi \text{ rad}} \times \frac{180^\circ}{1^\circ} \times \frac{60 \text{ arcmin}}{1^\circ} \times \frac{60 \text{ arcs}}{1 \text{ arcmin}} = \boxed{0.0422 \text{ arcseconds}}$$

26. An electron travels with speed $4.8 \times 10^6 \frac{\text{m}}{\text{s}}$ at an angle of 36° to a uniform magnetic field. If its path has a circular component of radius 0.075m, calculate the magnitude of the magnetic field.

ANSWER: $2.142 \times 10^{-4} \text{ T}$



$$v_{\perp} \sin \theta = \frac{v_{\perp}}{v} \rightarrow v_{\perp} = v \sin \theta$$

$$r = \frac{mv_{\perp}}{qB} \rightarrow B = \frac{mv \sin \theta}{rq} = \frac{(9.11 \times 10^{-31} \text{ kg})(4.8 \times 10^6 \frac{\text{m}}{\text{s}}) \sin(36^\circ)}{(0.075 \text{ m})(1.6 \times 10^{-19} \text{ C})} = \boxed{2.142 \times 10^{-4} \text{ T}}$$

$$\frac{\text{kg}}{\text{C}\cdot\text{s}} = \frac{\text{N}\cdot\text{s}}{\text{C}\cdot\text{m}} = \frac{\text{kg}\cdot\text{m}\cdot\text{s}}{\text{s}^2\cdot\text{C}\cdot\text{m}}$$

27. What is the maximum strength of the magnetic field in an electromagnetic wave that has a maximum electric-field strength of $1.9 \times 10^3 \text{ V/m}$?

ANSWER: $6.333 \times 10^{-6} \text{ T}$

$$c = \frac{E}{B} \rightarrow B = \frac{E}{c} = \frac{1.9 \times 10^3 \frac{\text{N}}{\text{C}}}{3 \times 10^8 \frac{\text{m}}{\text{s}}} = \boxed{6.333 \times 10^{-6} \text{ T}}$$

+4

28. A particular Van de Graaff generator has a metal sphere of diameter 14. cm that produces a voltage of 219. kV near its surface. What excess charge resides on the sphere?

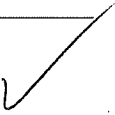
ANSWER: _____

5

29. Calculate the power dissipated by a car headlight that draws a current of 5.0 A when it has a voltage applied to it of 13. V.

ANSWER: 65 W

$$P = IV = (5 \text{ A})(13 \text{ V}) = \boxed{65 \text{ W}}$$



30. Calculate the critical angle for light traveling in a material that is surrounded by air, given that the material has an index of refraction of 1.3.

ANSWER: _____

6

+1